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DRAM Failure Prediction with Correctable Error Spatial Patterns: A Hybrid Learning Approach

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ABSTRACT

Dynamic Random Access Memory (DRAM) failures are a leading cause of server downtime in cloud centers. Predicting DRAM failures could reduce downtime events, avoiding data loss and industry damage. In this study, a novel DRAM failure prediction method based on spatial patterns of Correctable Errors (CEs) was proposed. After reconstructing the CE information into a feature map, effective artificial features were extracted. And an innovative 3D CWT-PCA transform was proposed to render its spatial patterns. We explored feature importance using SHAP analysis and visualisation techniques. Then, ResNet and KAN networks were employed for hybrid learning to predict DRAM failures. After achieving promising results on the publicly available Large-scale Dataset, we further collected real-world cloud center data for generalization validation.

KEYWORDS

DRAM Failure, Prediction, AIOps, Hybrid Learning, Correctable Error

1 INTRODUCTION

In the era of cloud computing, cloud servers have been widely deployed and adopted across various domains. In large-scale cloud data centers, Dynamic Random Access Memory (DRAM) failures constitute a significant portion of hardware malfunctions, accounting for approximately 1/3 of all server hardware failures [1]. Prior to the occurrence of DRAM failures, memory errors including Correctable Errors (CEs) and Uncorrectable Errors (UEs) are frequently observed. Modern computer systems incorporate Error Correction Code (ECC) mechanisms to autonomously detect and correct data corruption errors [2]. However, when the number of erroneous bits becomes excessive or the errors span across multiple memory chips, they may surpass the correction capability of ECC, potentially resulting in DRAM failures [3]. Moreover, the ECC implementation on modern Intel platforms is relatively less robust, making it more susceptible to bit errors originating from a single memory chip [4]. Therefore, relying solely on ECC is insufficient to ensure the reliability of DRAM systems.

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Currently, using CE information to predict DRAM failures has become a widely accepted approach [4]. In addition, various hardware information (such as system architecture, vendor, and DIMM model), operational logs (such as temperature and workload), and out-of-band logs (such as SEL and iBMC black box logs) are also commonly utilized [5-8]. In CE-based research, CE information was typically processed as tabular features and analyzed using Machine Learning (ML) methods [9-16]. As research progressed, the spatial patterns of CEs received increasing attention and were shown to outperform traditional tabular features [17, 18]. However, previous studies did not fully explore the effectiveness of CE spatial features for DRAM failure prediction.

This study focuses on CE spatial patterns, thoroughly explores various features, and develops a novel DRAM failure prediction method. To facilitate further research and improve reproducibility, the algorithm proposed in this study is publicly available on GitHub (https://github.com/syhnndas/dram_failure_prediction). The contributions of this study are as follows:

- We novelly reconstructed the CE information into a 3D feature map and extracted distance and area metrics.
- A unique 3D CWT-PCA transform was proposed to render the feature maps.
- A hybrid learning framework was constructed to integrate handcrafted and deep features.

2 RELATED WORK

DRAM failure prediction methods can be broadly categorized into three classes: those based on hardware runtime characteristics, CE tabular features, and CE spatial features. In hardware runtime-based methods, Sun et al. [5] extracted static parameters such as memory speed and memory age to predict DRAM failure. Mukhanov et al. [6] analyzed hardware features such as memory access rate and cache miss rate, and compared various supervised

learning methods. Giurgiu et al. [7] extracted event logs and sensor parameters, and constructed an integrated learning approach. Literature [8] proposed a coding methodology called Hierarchical DRAM Error Code (HiDEC), which analyzed the value of workloads for DRAM failure prediction from both macro and micro perspectives. However, hardware runtime characteristics often fall short in sufficiently characterizing DRAM faults.

In studies based on CE tabular features, Du et al. [9] proposed an online learning approach to process CE information and predict DRAM faults. Alibaba Cloud hosted a DRAM failure prediction challenge at PAKDD 2021. Using the released dataset, Wu et al. [10] conducted a comparative study on the performance of classical ML models. Subsequently, Alibaba conducted an in-depth study on CE information, including bank, rank, row, and column, and released a larger dataset [11]. Bogatinovski et al. [12] input the feature vectors of CE location fields into XGBoost for DRAM failure prediction. Research [13, 14] further analyzed the location, number, and frequency features of CE, and conducted experiments using various classifiers. Studies [15, 16] analyzed statistical information such as the difference in the number of CEs across different banks. They also explored the correlation between location fields and faults. Although these studies explored CE features in depth, they were limited to tabular features and traditional machine learning approaches.

In recent years, the spatial patterns of CEs have attracted much attention for their excellent prediction effects. Typical research comes from Intel [17], where scholars revealed several high-risk CE alignment patterns and performed DRAM failure prediction based on them. Study [18] also explored the method of predicting UE based on CE spatial structure. And They proved the effectiveness of the CE spatial structure. However, current researches on prediction based on spatial structure are immature and lack of systematic algorithms. To address these shortcomings, this study proposes a hybrid learning approach based on CE spatial patterns.

3 METHODS

3.1 CE Space Reconstruction

In the microarchitecture of DRAM, each DRAM module consists of multiple Dual-Inline Memory Modules (DIMMs). Each DIMM has two ranks, and each rank contains 16 banks. Each bank has 2^{17} rows and 2^{10} columns [11]. Therefore, the CE log contains fields such as rank, bank, row, and col to identify the location of errors, as shown in the table at the top of Fig. 1(a).

In this study, we reconstructed the bank, row, and column fields into multi-channel feature maps. Specifically, we treated the three fields as a 3D coordinate [*bank*,*row*,*col*], and assigned a value of 1 (gray square) at that position to indicate the occurrence of one CE. All other positions were set to 0 (white squares). This process resulted in the feature map shown in Fig. 1. Considering the maximum values of bank, row, and col, we set the feature map dimensions to $Row = 2^{17}$, $Col = 2^{10}$, and $Bank = 16$.

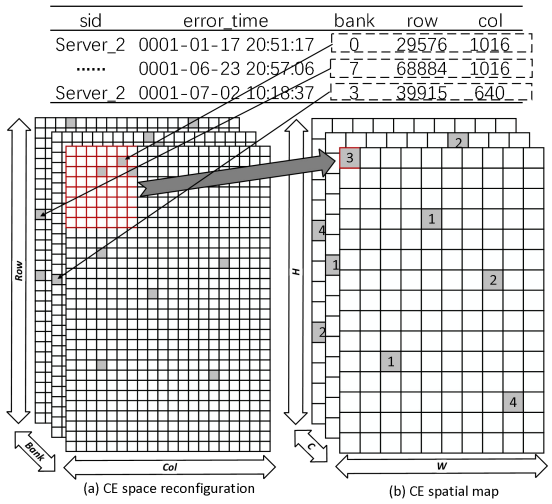


Fig. 1. CE Spatial Reconfiguration Methods

Considering that the feature map may contain many “zeros”, which is unfavorable for feature aggregation, we performed a simple compression operation. Specifically, the original map in Fig. 1(a) was compressed into a smaller $H \times W \times C$ map, as shown in Fig. 1(b). The numbers of the corresponding regions were superimposed and the regions were not overlapped during compression. The resulting CE spactial map was used for subsequent feature extraction.

3.2 Artificial Feature Extraction

Studies [17, 18] demonstrated that the more dispersed the distribution of CEs, the more likely it may be to lead to UEs and DRAM failures. Meanwhile, study [15] concluded that the difference in the number of CEs across banks is a good predictor. Based on these findings, we extracted 10 Artificial features (AFs) from the CE spactial map, as shown in Table 1. The schematic diagram of the extraction method is shown in Fig. 2.

The features *HD*, *WD*, *CD*, *2DD*, *3DD* all measure the distance between two CE clusters, and the concise formulas have been attached to the figure. And the metrics *MHD*, *MWD*, *MCD*, *M2DD*, *M3DD* are the corresponding maximum values to measure the degree of CE dispersion. *NA16*, *NA64*, *NA144* are the maximum number of CEs covered by different areas on a particular bank, which measures the degree of CE clustering on this bank.

Table 1. List of artificial features

Features	Description
NCE	Number of CE
MHD	Maximum Height Distance
MWD	Maximum Width Distance
MCD	Maximum Channel Distance
NCEE	Number of CEs on the edges
M2DD	Maximum 2D distance
M3DD	Maximum 3D distance
NA16	Number of CEs covered by an area of 16
NA64	Number of CEs covered by an area of 64
NA144	Number of CEs covered by an area of 144

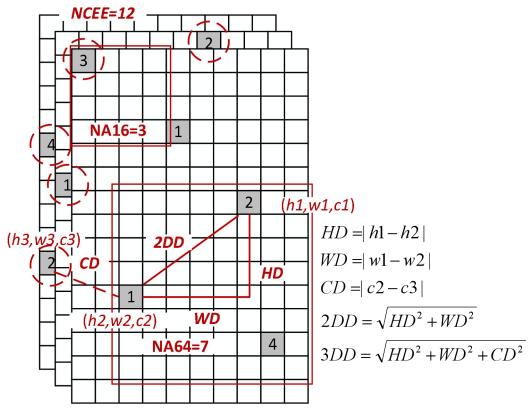


Fig. 2. Schematic of the feature extraction process

3.3 3D CWT-PCA Transform

We proposed a unique transform method to extract wavelet coefficients from CE spactial map, and then downscaled it to RGB image. For the 3D matrix $X(h,w,c) \in R^{H \times W \times C}$, discrete CWT was performed for each row h of each channel c:

$$W(h,w,a,c) = \sum_{t=1}^{W} X(h,t,c)\psi^{*}_{a,b}(t) \tag{1}$$

where a is the scale and $b = w$ is the time position. The wavelet basis function $\psi(t)$ is the Morlet wavelet, $\psi_{a,b}(t)$ is the result after scale transformation:

$$\psi_{a,b}(t) = \frac{1}{\sqrt{|a|}}\psi\left(\frac{t-b}{a}\right) \tag{2}$$

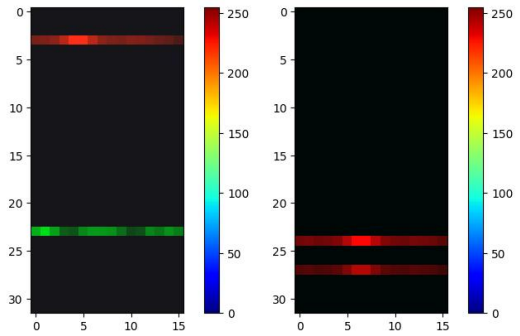
Then, take the mean on scale a:

$$CWT_{mean}(h,w,c) = \frac{1}{A}\sum_{a=1}^{A} |W(h,w,a,c)| \tag{3}$$

The coefficient matrix $CWT_{mean}(h,w,c)$ of shape [H,W,C] was then obtained. Since the number of channels C may be large, it is not conducive to feature aggregation. We further perform PCA dimensionality reduction on $CWT_{mean}(h,w,c)$ to compress C into C'. This step was performed using the PCA function of the scikit-learn package. And the pixel values were linearly normalized to [0,255]. The final coefficient map obtained was called *CWTmap* and the shape was [H,W,C']. The concise steps are as follows:

$$CWT_{mean}(h,w,c) \xrightarrow{PCA} CWT_{mean}(h,w,c') \xrightarrow{Normalization} CWTmap(h,w,c') \tag{4}$$

In this article, several comparison experiments were conducted and finally H=32, W=16, C=16 and $C' = 3$ were selected. The *CWTmap* obtained from the two example samples is shown in Fig. 3. These images will be used for the deep learning network.



(a) Sample Server_23495 (b) Sample Server_27867

Fig. 3. *CWTmap* generated by 3D CWT-PCA method

3.4 Hybrid Learning

We constructed a ResNet network [19] to extract features from the *CWTmap*, which consisted of four residual blocks. Initially, the complete ResNet architecture, including the final Linear and Softmax layers, was used for training. After training, these two layers were frozen, and the output from the Flatten layer was extracted as Deep Features (DFs). Following the fusion of Artificial Features (AFs) and DFs, SHAP analysis [20] was applied for feature selection. In addition, the KAN network [21] had the advantage of learnable activation functions. And it can replace the weight parameters with learnable univariate functions. We introduced the KAN network for final prediction.

4 EXPERIMENTS

4.1 Dataset

We used the Large-scale Dataset released by Alibaba Group [11]. It contained historical data from 30,502 servers. The dataset included over 75 million CE records, collected over 8 months. During this period, 3,017 servers experienced failures. Among them, 2,139 servers had CEs recorded before the failure. The dataset included three files, “mcelog” recorded CE logs, including timestamp, rank, bank, row, and column; “inventory” provided hardware vendor and model information; “trouble_tickets” contained server IDs and failure times. To ensure class balance, we used random undersampling [22] to select the same number of normal and failure samples. The data was then split into training and testing sets in an 8:2 ratio. Following [10], our prediction experiment was performed 5 minutes in advance, i.e., only CE records generated more than 5 minutes before the failure were used.

In addition, real-world production data were collected from the Inspur Cloud Data Center. More than 3,600 servers across 14 test cloud centers nationwide were monitored. From July 2023 to March 2025, over 1 million CE records were collected. A total of 106 servers experienced DRAM failures, all of which emitted CEs before the failure. Fig. 4 shows the CE information fields, which included key information such as bank (ba), row (ce_row), and col (ce_col). Random undersampling was again used to select 220 normal samples. Due to the smaller data size, the Inspur Cloud

Data was only used to evaluate the generalization ability of the proposed method. Table 2 summarizes all datasets used in this study, along with the number of samples and their purposes.

id	datetime	node	channel	slot	socket	ba	ce_row	ce_col
1217113	2025-03-18 08:53:24	compute195	2	1	1	2	1ec69	3d0
1217107	2025-03-18 05:14:56	compute130	0	0	1	0	bce	1e0
1217101	2025-03-17 21:11:29	compute71	0	0	0	3	a4d7	1e0
1217098	2025-03-17 17:24:05	compute19	1	0	0	0	e2f1	1d8
1217095	2025-03-17 14:25:01	compute18	1	0	0	3	42c	350
1217093	2025-03-17 14:24:11	compute16	1	0	0	3	42c	350
1217088	2025-03-17 09:29:24	compute139	2	1	0	3	1c692	280
1217081	2025-03-16 23:01:50	compute195	2	1	1	2	1ec69	3d0
1217078	2025-03-16 21:42:29	compute71	0	0	0	3	a4d7	1e0
1217073	2025-03-16 11:15:52	cmp060	0	1	1	0	de5b	70
1217063	2025-03-16 02:02:54	cmp060	0	1	1	0	de5b	70
1217061	2025-03-16 00:17:31	cmp108	0	0	1	1	1e21	3a8
1217057	2025-03-15 22:12:27	compute71	0	0	0	3	a4d7	1e0
1217054	2025-03-15 18:00:03	compute195	2	1	1	2	1ec69	3d0
1217046	2025-03-15 08:16:55	cmp060	0	1	1	0	de5b	70
1217044	2025-03-15 07:51:19	cmp060	0	1	1	0	de5b	70
1217041	2025-03-15 07:24:42	cmp060	0	1	1	0	de5b	70
1217036	2025-03-15 01:54:26	compute195	2	1	1	2	1ec69	3d0
1217032	2025-03-14 22:42:25	compute71	0	0	0	3	a4d7	1e0
1217029	2025-03-14 19:36:55	compute17	2	1	0	2	c5	0

Fig. 4. CEs in the Inspur Cloud Data

Table 2. Datasets, sample sizes, and their usage

Dataset	Number of samples (Failure / Normal)	Usage
Large-scale Dataset	4278 (2139/2139)	Training and testing
Inspur Cloud Data	326 (106/220)	Generalizability validation

4.2 Experiments

The overall experimental workflow is illustrated in Fig. 5. On one hand, the CE spatial map was transformed using the 3D CWT-PCA method to generate the *CWTmap*, which was then fed into the ResNet for feature extraction. The difference between ResidualBlock-1 and ResidualBlock-2 lay in the padding settings of their convolutional layers. By configuring the Flatten layer to output 16 dimensions, a total of 16 Deep Features (DFs) were extracted, denoted as *DF1* to *DF16*.

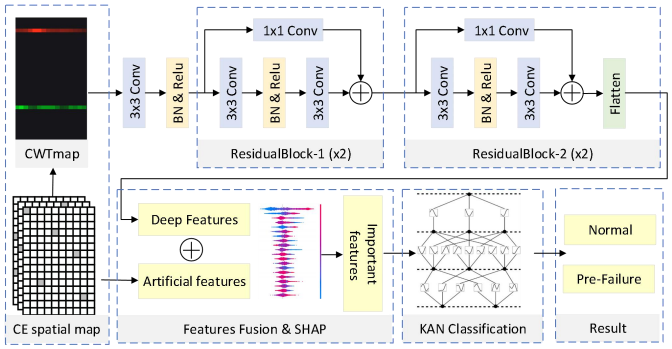


Fig. 5. Experimental Workflow

On the other hand, a set of Artificial Features (AFs) was extracted from the CE spatial map, as listed in Table 1. These handcrafted features were then combined with the DFs. SHAP analysis [20] was applied to identify the most important features. Finally, the KAN network was used to predict whether a DRAM failure would occur in the near future.

5 RESULTS AND ANALYSIS

5.1 SHAP Analysis

After the fusion of DF features and AF features, a total of 26 features were obtained. The results of the SHAP analysis on these features are shown in Fig. 6. The x-axis represents the SHAP values, which describe the impact of each feature on the model's output. Larger positive or negative values indicate greater importance. The y-axis represents the feature identifiers. Only the top 20 most important features are shown in the figure. The order of the features on the y-axis is based on the absolute value of the SHAP values, with the most important features listed at the top.

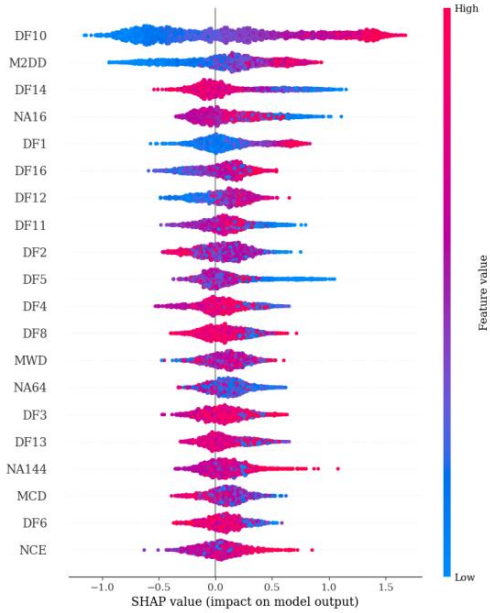


Fig.6. SHAP analysis results

As shown, among the top 10 most important features, 8 are DF features, demonstrating the feature extraction capability of the ResNet. *M2DD* and *NA16* are relatively important AF features, both extracted from the same bank. Similarly, *MWD*, *NA64*, and *NA144* are features from the same bank. This suggests that the CE distribution characteristics on the same bank have strong potential for DRAM fault characterization. In contrast, *NCE*, as well as *MHD*, *NCEE*, and *M3DD*—features not shown in the figure—were relatively less important. Through multiple comparative experiments, the optimal feature set was selected. The best prediction performance was achieved using the top 18 features.

5.2 DRAM Failure Prediction Results

We conducted prediction experiments using only DF features, only AF features, and the fusion of both. The impact of SHAP was also discussed. When SHAP was applied, the top 18 most important features were selected for the experiments. The results on the Large-scale Dataset are shown in Table 3. In the calculation of metrics such as Recall and Precision, failure samples were considered as positive samples, and normal samples were considered as negative samples.

Table 3. DRAM failure prediction results (on Large-scale Dataset)

Features	Result (%)			
	Accuracy	Recall	Precision	F1-score
DF	71.23	75.43	68.59	71.85
DF+SHAP	79.65	76.43	70.32	73.25
AF	69.87	66.78	72.12	69.35
AF+SHAP	70.33	69.18	75.64	72.27
DF+AF	**82.30**	77.23	**76.22**	76.72
DF+AF+SHAP	81.60	**79.73**	74.88	**77.23**

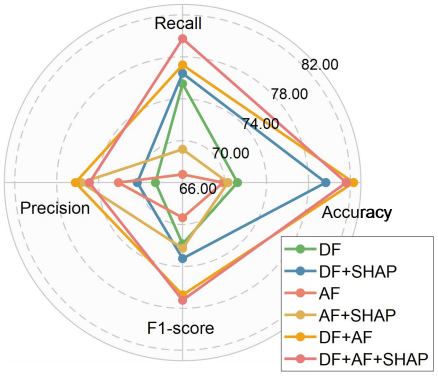


Fig. 7. Results of different feature combinations

A donut chart was plotted to further compare the performance of various feature combinations, as shown in Fig. 7. In such failure prediction experiments, the Recall metric is relatively important. When only AF features were used, the Recall was 66.78%, indicating that the handcrafted features extracted were insufficient to represent DRAM faults. In contrast, DF features performed better than AF features. The best performance was achieved when both feature sets were used together, with a Recall of 77.23%. The importance of the SHAP algorithm is also evident. Whether using AF or DF alone, or both in combination, incorporating SHAP consistently improved performance. The DF+AF+SHAP combination achieved the highest Recall and F1-score.

5.3 Effects of Different Classifiers

Experiments were conducted on Random Forest (RF), XGBoost, and the KAN network, with results shown in Fig. 8. The dataset used was the Large-scale Dataset. The KAN network, an improvement on MLP [21], was proposed in 2024. The results showed that RF achieved the highest Accuracy, XGBoost achieved the highest Precision and F1-score. And the KAN network achieved the highest Recall, with an acceptable F1-score, making it the best-performing model.

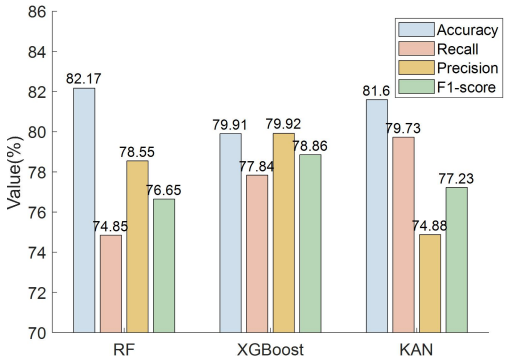


Fig. 8. Effects of different classifiers

5.4 Generalizability Verification

After developing the method on the public dataset, we conducted generalization experiments on data from real-world cloud center. The parameters of the ResNet and KAN models were determined based on the public dataset. The entire pipeline was then applied to the Inspur Cloud Data. And forecasts were made for different time windows in advance. Different time windows mean different amounts of CE data are available. As shown in Fig. 9, assuming that a DRAM failure occurs at the moment t_l and the prediction window is T, only the data before the moment t_0 is available.

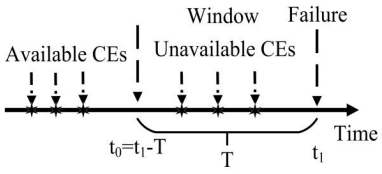


Fig. 9. Schematic of the prediction time window

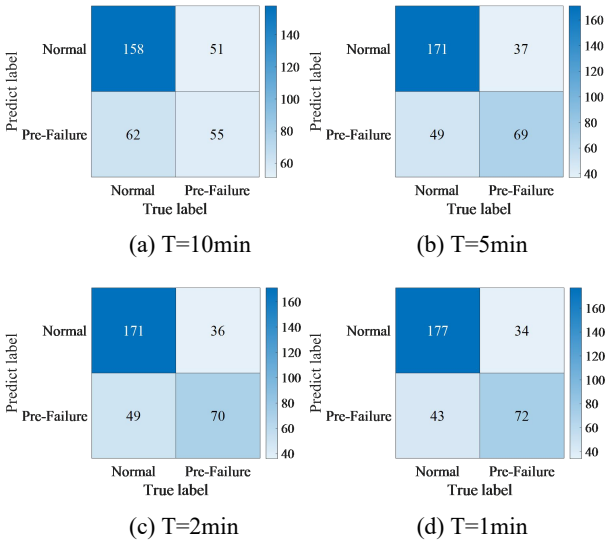
Table 4. Prediction results on the Inspur Cloud Data.

Prediction window (T)	Accuracy	Recall	Precision	F1-score
10min	65.34%	51.89%	47.01%	49.33%
5min	73.62%	65.09%	58.47%	61.61%
2min	73.93%	66.04%	58.82%	62.22%
1min	76.38%	67.92%	62.61%	65.16%

In the Large-scale Dataset, predictions were made with a 5-minute time window. To examine the impact of the prediction window, experiments were conducted on the real cloud datacenter data with time windows of 10 minutes, 5 minutes, 2 minutes, and 1 minute, respectively. The results are shown in Table 4. When predicting with a 10-minute lead time, the Recall was only 51.89%. As it gets closer to the failure, the Recall value increases. This is similar to the findings in [11], where a large number of CEs often occur just before a DRAM failure.

Currently, the typical time required to mitigate DRAM failure in the industry is around 5 minutes [10], making 5 minutes the ideal prediction window. However, the results for 2-minute and 1-minute lead times indicate that shorter prediction windows yield better prediction performance. This also suggests that industries should strive to reduce the failure mitigation time. As shown in the

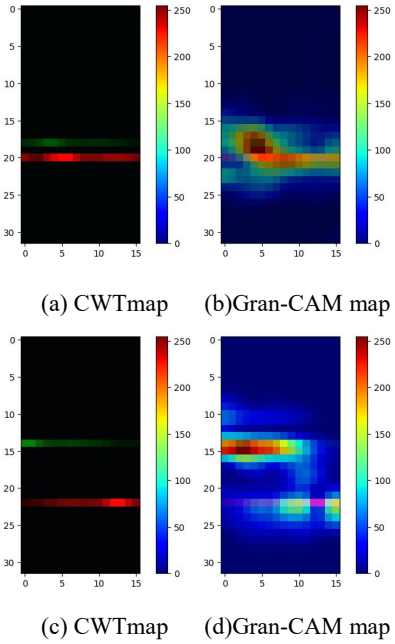
confusion matrix in Fig. 10, the number of predicted Pre-Failure samples increases as the prediction window shortens (from 55 to 72).



(a) T=10min (b) T=5min

(c) T=2min (d) T=1min

Fig. 10. Confusion matrix for different prediction windows



(a) CWTmap (b)Gran-CAM map

(c) CWTmap (d)Gran-CAM map

Fig. 11. Grad-CAM Visualization

Table 5. Comparison of Related Studies

Literatures	Focus	Specific features	Result (%)		
			Precision	Recall	F1-score
X. Du et al. [9], 2018	CE tabular features	Cells, Rows, and Columns	64.9	57.1	60.3
Z. Wu et al. [10], 2021		Rows, Columns and Kernel address features	58.9	40.5	48.0
Z. Cheng et al. [11], 2022		Counter features, Component features, Statistical features and Configuration features	59.6	66.8	62.9
Q. Yu et al. [14], 2023		CE Error Rate, Fault Counts, DQ-Beat Error bits, Error Bit Patterns and Memory Events	63.5	57.4	60.3
F. Yu et al. [15], 2021		Address mce features and Kernel address features	50.9	63.6	56.6
C. Li et al. [17], 2022	CE Spatial patterns	Risky CE Indicator and Fault indicators	**78.0**	81.5	79.7
Q. Yu et al. [18], 2023		DQ-Beat Error Bits and CE pattern indicators	53.2	**82.0**	64.4
Our work		**Artificial features, deep features, hybrid learning in CE spatial patterns**	**74.88**	**79.73**	**77.23**

5.5 Visualization Exploration

SHAP analysis clearly identifies which artificial features are relatively important. However, the black-box nature of deep learning makes it challenging to determine the importance of deep features. In this section, the Grad-CAM algorithm was used to explore the importance of the *CWTmap* and ResNet. The visualization results for two randomly selected *CWTmaps* are shown in Fig. 11.

It can be observed that the network focuses more on the regions with low wavelet coefficient energy (green areas) in the CWTmap, while giving less attention to high-energy regions (red areas). This may be because the low-energy regions correspond to more scattered CE distributions. This indirectly suggests that scattered CEs are more likely to lead to DRAM failures. However, this conclusion needs further validation through subsequent experiments.

6 DISCUSSION

The studies on DRAM failure prediction are mainly based on CE table features, while there are also some studies based on spatial structure. The latest related studies are shown in Table 5. Most of the primitive table features, such as Cells, Rows, Columns and Kernel address, do not have very good failure characterisation capability, with the resulting Recalls of only 57.1% and 40.5%. Later, scholars proposed a variety of derived features based on the original features, such as CE Error Rate, DQ-Beat Error bits and so on. These features enhanced Recall and confirmed the effectiveness of feature engineering. However, these researches were still not free from the limitations of form features.

In contrast, the discovery of spatial features, especially the Risky CE pattern [17], has pushed the research in this field forward. It seems that the spatial layout of the CE can approximate the representation of the error bits in the actual memory stick where they occur. They obtained the Recall of 78.0%. Similarly, study [18] inherited this finding and combined it with DQ-Beat Error Bits from [14] to obtain better results. However, the studies [17,18] do not seem to be suitable for engineering landing, reflected in the idealistic and not easily quantifiable nature of the Risky CE pattern. As well as the fact that DQ-Beat Error bits are difficult to compute in real-time in large cloud centres, they are only suitable for analysing historical data. In addition, their study only used publicly available Large-scale Dataset and did not verify the feasibility on real cloud centre data.

This study drew on the ideas of [17, 18] and, based on this, innovatively proposed to reconstruct the CE into a feature map for quantification. In addition a wide range of features containing both artificial and deep features are extracted, making the features more comprehensive compared to previous studies. Comparable results to existing studies were obtained on the public dataset, i.e., the Recall of 79.73% and the Precision of 74.88%. The proposed method has been successfully engineered and deployed on a large-scale cluster at the Inspur Cloud Center, yielding a Recall of 65.09% as reported in Table 4.

7 CONCLUSION

This study focused on the CE spatial patterns and delved into various features to establish a novel DRAM failure prediction method. We reconstructed CE information into feature maps, and proposed a 3D CWT-PCA transformation for rendering them. The extracted artificial and deep features were fused and selected. Finally, a hybrid learning approach using ResNet and KAN networks was employed for DRAM failure prediction. Feature selection and visualization techniques were used to uncover important CE spatial features and patterns. The proposed method achieved good results on the public Large-scale Dataset, and demonstrated strong generalization on real-world cloud center data. These contributions would provide valuable insights for AIOps tasks in cloud centers.

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